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> *Confidential DD dossier. Status tags:* ***BUILT+PROVEN / DESIGNED / RESEARCH-DIRECTION****. Everything below is grounded in the* `arc-lean` *source tree; file names are real and readable. Foundation:* `THESIS-collective-nonparametric-intelligence.md`*.*

The shape of the thing

Aiko is a single native binary (Zig, zero external dependencies) that is at once: a **contract substrate** (an append-only ledger of intents and verdicts), an **energy-based harness** (a non-parametric router over that ledger), and a **client** to a live index of the world (Unbrowse) and to Solana. There is no Python, no model server, no data files: the embedding is baked in, the "weights" are reconstructible from the ledger, and the reasoning model is grounded through the harness rather than baked into it.

1. The atom — a contract neuron

The unit of everything is a **contract neuron**: one append-only JSONL row recording an intent (a *declare*) and its lifecycle (satisfied / died / purged — the terminal alphabet is exactly those three). A neuron carries its action, its lineage signature (every row is signed from a single wallet-derived root key — `lineage.zig`), and its pointer type. Contracts compose into a graph via `parent_id` containment and `contract:`-ref cross-edges. **BUILT+PROVEN** (`contract.zig`, `ledger.zig`, `pointer.zig`, `terminal.zig`).

The ledger is the shared, auditable memory. It is not a log *of* the system — it *is* the system's state, and the model is reconstructed from it.

2. The energy model — non-parametric, ~1 KB, trained on-fire

The harness's model is deliberately **not** a big parametric net. It is two thin, cheap primitives:

- **Embedding (**`embed.zig`**)** — a deterministic feature-hashing embedding (the hashing trick): each token and character-trigram hashes into a fixed-dimension, L2-normalized vector. Baked into the binary — zero data files, zero network. Cosine over these vectors captures lexical + sub-word overlap. **BUILT+PROVEN.**
- **Energy prototype (**`ebm.zig`**)** — a `Prototype` is a **reward-weighted running-mean centroid** over context vectors (West 1979; O(1) per fire). `foldText(action, reward)` folds a firing in: positive reward pulls the centroid toward it (Hebbian potentiation), negative pushes away (inhibition — the contrastive negative phase). The **energy** of a context is its negative cosine to the prototype; routing is **argmin energy**. **BUILT+PROVEN.**

Each such head is ~1 KB and reconstructible from the ledger. **The knowledge is not in the weights** — it is in the retrievable index (the world) plus the collective firing history (the ledger). A thought costs a retrieval + a centroid fold, not a forward pass through a giant net.

3. Fractal keys — multi-scale priors for free

The same firing folds into the *same atom* at many scales at once — `invoke:{event}`, `:L{layer}`, `:dim:{w6h}`, `:verb:`, `:ptype:`, `:root`. One cheap operation yields a **coarse-to-fine router**: the collective can prioritize at every granularity — per-event, per-layer, per-dimension — **without a separate model per scale.** This is the "fractal" in the name: the EBM is not one model but many nested thin ones, each wrapping a different facet of a route, broken down to first principles. **BUILT/DESIGNED** (fractal key convention in `fractal.zig` / `pattern.zig`; the arc-lean router exercises the per-run centroid directly).

4. Energy-ranked resolution — the router IS the harness

This is the load-bearing mechanism and the falsifiable atom of the whole thesis. When the substrate has a frontier of candidate contracts to fire next, it does **not** scan them in arbitrary order. `energyRankCandidates` (in `main.zig`) reconstructs a `Prototype` from the run's resolved history (each satisfied action folded with reward +1, each died with −1 — the train-on-fire signal, the ledger *is* the weights), then **sorts the frontier best-first by energy** so the most "resolved-like" candidates fire earliest — shorter cascades, less wasted work.

It is **always on** and **reorder-only**: every candidate still fires, so nothing is starved (a fairness floor holds by construction). It is the **thinking-budget allocator** — it spends scarce compute/spend where history says it pays. **BUILT+PROVEN**, and *measured*: on the prove-or-kill benchmark it settles a producer/consumer graph in **2 passes vs 3** for scan-order, mutation-verified (reversing the sort restores 3 — so the ranking, not luck, is doing the work). This is the atom "the collective routes its own thinking," proven before the cathedral. (See `04-PROOF-traction.md`, `PLAN-contracts-as-onchain-ebm-weights.md`.)

5. Ledger-reconstructible weights — the model is the commons

Because every weight is a fold of ledger rows, **anyone can rebuild the prior from the shared ledger** (`replayFromLedger`, on `main`). The "weights" are the collective experience, in the open — forkable, inspectable, re-derivable. No single node is "the model": the graph of contracts + the ledger-reconstructed EBM

- the indexed world is the mind. This is also what makes the weights **portable on-chain**: a weight is just content-addressed ledger data (see `03-ONCHAIN-privacy-solana.md`, and `cas.zig` — content-addressed store, dedup-for-free, **BUILT+PROVEN**).

6. Grounding a small model — capability without parameters

The reasoning model is trained to be **as small as possible** (non-parametric in spirit: it holds instruction-following and reasoning, not world-knowledge). On its own it "knows little." But the harness **grounds** it: retrieval brings the relevant slice of the world and the machine to bear, and energy routing tells it which action history says works. A small model grounded this way is **smarter than the average model of the same footprint** — capability comes from grounding, not scale. **Status: the small model is DESIGNED / in-training; the grounding mechanism (retrieval + `embed`/`ebm` energy routing) is BUILT+PROVEN.** The leap from "thin router" to "reasoning-grade intelligence" is the honest open problem — stated plainly in the THESIS and `05-RISK-honest-wall.md`.

7. The module map (what's real, by concern)

Concern	Modules (`libcontract/src/`)	Status
Energy harness	`embed`, `bge` (baked BERT), `ebm`, `route` (`recommend`), `pattern`, `fractal`	**BUILT+PROVEN** (router measured — 88.9% LOO, `route_bench.zig`)
Contract substrate	`contract`, `ledger`, `pointer`, `terminal`, `cache`, `guards`, `capability`	**BUILT+PROVEN**
Identity / truth-root	`lineage`, `keystore`, `bible`, `verses`	**BUILT+PROVEN**
Commons economics	`cas`, `reward`, `usage`, `usage_ledger`, `epoch`	**BUILT+PROVEN** (local)
Solana / on-chain	`voltr`, `stake`, `payout`, `broadcast`, `broadcast_card`, `solana_rpc`, `solana_tx`, `base58`	**BUILT+PROVEN** (read/derive live; write = operator-gated)
Small reasoning model	(separate training track)	**DESIGNED / in-training**

Every module ships its tests *inside its source file*, exercised by `zig build test` and gated by `scripts/build.sh` (unit tests + 53 doctrine-parity claims + a signed self-attestation). Green build = every claim above has matching, tested code.